

MAP112 - Determinantes - Álgebra Linear

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1. Determinantes 2×2

Calcule os determinantes:

1a. $\begin{vmatrix} 3 & 5 \\ 2 & 4 \end{vmatrix} = (3 \cdot 4) - (5 \cdot 2) = 12 - 10 = 2$
 1b. $\begin{vmatrix} -1 & 4 \\ 6 & 2 \end{vmatrix} = (-1 \cdot 2) - (4 \cdot 6) = -2 - 24 = -26$
 1c. $\begin{vmatrix} 0 & 3 \\ 5 & 1 \end{vmatrix} = (0 \cdot 1) - (3 \cdot 5) = 0 - 15 = -15$

2. Regra de Sarrus

Calcule os determinantes utilizando a regra de Sarrus:

1a. $\begin{vmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 0 & 4 & 1 \end{vmatrix} = (1 \cdot 1 \cdot 1) + (0 \cdot 3 \cdot 0) + (2 \cdot 2 \cdot 4) = 1 + 0 + 16 = 17$
 $(2 \cdot 1 \cdot 0) + (1 \cdot 3 \cdot 4) + (0 \cdot 2 \cdot 1) = 0 + 12 + 0 = 12$
 $17 - 12 = 5$

1b. $\begin{vmatrix} 2 & 1 & 0 \\ 1 & 3 & 4 \\ 5 & 2 & 1 \end{vmatrix} = (2 \cdot 3 \cdot 1) + (1 \cdot 4 \cdot 5) + (0 \cdot 1 \cdot 2) = 6 + 20 + 0 = 26$
 $(1 \cdot 3 \cdot 5) + (2 \cdot 4 \cdot 2) + (1 \cdot 1 \cdot 1) = 0 + 16 + 1 = 17$
 $26 - 17 = 9$

3. Memória e Regras

Dada a matriz:

$$A = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 2 & 4 & 0 \end{vmatrix}$$

calcule:

10. M_{11} e C_{11}

1	2	3
0	1	2
2	1	0

$$\begin{aligned} \text{Minor} &= (1 \cdot 0) - (1 \cdot 2) \\ &= 0 - 2 \\ &= -2 \end{aligned}$$

$$\begin{aligned} \text{Cofactor} &= (C_{11}) = (-1)^{1+1} \\ &= (-1)^2 \cdot (-2) \\ &= 1 \cdot (-2) \\ &= -2 \end{aligned}$$

11. M_{23} e C_{23}

1	2	3
0	1	2
2	1	0

$$\begin{aligned} \text{Minor} &= (1 \cdot 1) - (2 \cdot 2) \\ &= 1 - 4 \\ &= -3 \end{aligned}$$

$$\begin{aligned} \text{Cofactor} &= (C_{23}) = (-1)^{2+3} \cdot (-3) \\ &= (-1)^5 \cdot (-3) \\ &= -1 \cdot (-3) \\ &= 3 \end{aligned}$$

12. M_{32} e C_{32}

1	2	3
0	1	2
2	1	0

1	3
0	2

$$\begin{aligned} \text{Minor} &= (1 \cdot 2) - (3 \cdot 0) \\ &= 2 - 0 \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{Cofactor} &= (C_{32}) = (-1)^{3+2} \cdot 2 \\ &= (-1)^5 \cdot 2 \\ &= -1 \cdot 2 \\ &= -2 \end{aligned}$$

13. Sistema de Laplace

Calcule o determinante da matriz abaixo:

$$A = \begin{vmatrix} 2 & 0 & 1 \\ 1 & 3 & 0 \\ 2 & 0 & 5 \end{vmatrix} = A$$

1a. Expandindo pela primeira linha

2	0	1
1	3	0
2	0	5

$$2 \cdot \text{cofator}_{11} + 0 \cdot \text{cofator}_{12} + 1 \cdot \text{cofator}_{13}$$

$$\text{Memor} = (3 \cdot 5) - (0 \cdot 0) \quad \text{Memor} = (3 \cdot 4) - (1 \cdot 0)$$

$$= 15 - 0 \quad = 12 - 0$$

$$= 15 \quad = 12$$

$$30 - 12 = 18$$

$$\text{cofator} = 2 \cdot (-1)^{1+1} \cdot 15 \quad \text{cofator} = 1 \cdot (-1)^{1+3} \cdot 12$$

$$= 2 \cdot (-1)^2 \cdot 15 \quad = 1 \cdot (-1)^4 \cdot 12$$

$$= 2 \cdot 1 \cdot 15 \quad = 1 \cdot 1 \cdot 12$$

$$= 30 \quad = 12$$

2a. Expandindo pela segunda coluna

2	0	1
1	3	0
2	0	5

$$0 \cdot \text{cofator}_{12} + 3 \cdot \text{cofator}_{22} + 0 \cdot \text{cofator}_{32}$$

$$\text{Memor} = (2 \cdot 5) - (1 \cdot 4) \quad \text{cofator} = (-1)^{2+2} \cdot 6$$

$$= 10 - 4 \quad = 1 \cdot 6$$

$$= 6 \quad = 6$$

$$3 \cdot 6 = 18$$

5. Propriedades - analise sem cálculos diretos

Determine o valor do determinante justificando com propriedades:

1a.

1	2	3
2	4	6
3	6	9

= Quando uma matriz se encontra linha múltipla de
de outra linha, ou coluna múltipla de outra coluna.
 $\rightarrow \det(A) = 0$

1b. $\begin{vmatrix} 0 & 0 & 0 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix}$ Linha ou coluna nula $\Rightarrow \det(A) = 0$

1c. $\begin{vmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{vmatrix}$ Duas linhas (ou colunas) iguais $\Rightarrow \det(A) = 0$

6. Operações Elementares

Calcule os determinantes utilizando operações elementares:

1a. $\begin{vmatrix} 1 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 1 & 1 \end{vmatrix}$ $\begin{vmatrix} 1 & 2 \\ 2 & 3 \\ 1 & 1 \end{vmatrix}$ $(1 \cdot 3 \cdot 1) + (2 \cdot 1 \cdot 1) + (1 \cdot 2 \cdot 1) = 3 + 2 + 2 = 7$
 $(1 \cdot 3 \cdot 1) + (1 \cdot 2 \cdot 1) + (2 \cdot 2 \cdot 1) = 3 + 2 + 4 = 9$
 $7 - 9 = -2$

$L2 \rightarrow L2 - 2L1$ $L3 \rightarrow L3 - L1$ $A' = \begin{vmatrix} 1 & 2 & 1 \\ 0 & -1 & -2 \\ 0 & -1 & 0 \end{vmatrix}$
 $= 2 \cdot 2 \cdot 1 = 0$ $= 1 \cdot 1 = 0$
 $= 3 - 2 \cdot 2 = -1$ $= 1 - 2 = -1$
 $= 4 - 2 \cdot 1 = 2$ $= 1 - 1 = 0$

$L3 \rightarrow L3 - L2$ $A' = \begin{vmatrix} 1 & 2 & 1 \\ 0 & -1 & -2 \\ 0 & 0 & -2 \end{vmatrix}$
 $= -1 \cdot (-1) = 1$ $= 0 - 2 = -2$

$(1 \cdot (-1) \cdot (-2)) + (2 \cdot 2 \cdot 0) + (1 \cdot 0 \cdot 0) = 2 + 0 + 0 = 2$
 $(1 \cdot (-1) \cdot 0) + (1 \cdot 2 \cdot 0) + (2 \cdot 0 \cdot (-2)) = 0 + 0 + 0 = 0$
 $2 - 0 = 2$

Adição de Múltiplo de Linhas (ou colunas)

De:

2	4	2	2	4	$(2 \cdot 3 \cdot 3) + (4 \cdot 4 \cdot 3) + (2 \cdot 4 \cdot 4) = 18 + 48 + 32 = 98$
1	3	4	1	3	$(2 \cdot 3 \cdot 3) + (2 \cdot 4 \cdot 4) + (4 \cdot 4 \cdot 3) = 18 + 32 + 48 = 98$
3	4	3	3	4	$98 - 98 = 0$

L2 $\rightarrow$ 2L2 - L1

$$= 2 \cdot 1 - 2 = 0$$

$$= 2 \cdot 3 - 4 = 2$$

$$= 2 \cdot 4 - 2 = 6$$

L3 $\rightarrow$ 2L3 - 3L1

$$= 2 \cdot 3 - 3 \cdot 2 = 0$$

$$= 2 \cdot 4 - 3 \cdot 4 = -2$$

$$= 2 \cdot 3 - 3 \cdot 2 = 0$$

A' =

2	4	2
0	2	0
0	2	0

L3 $\rightarrow$ L3 - L2

$$= 2 - 2 = 0$$

A' =

2	4	2
0	2	0
0	0	0

2 cofatores 11 + 0 cofatores 21 + 0 cofatores 31

Menor = $2 \cdot 0 - 0 \cdot 0$

Cofator = $2 \cdot (-1)^{1+1} \cdot 0$

$= 0 - 0$

$= 0$

$= 2 \cdot 1 \cdot 0$

$= 0$

Adição de Múltiplo de Linhas (ou colunas)

7. Matriz Triangular

Calcule os determinantes:

De:

2	0	0
1	3	0
4	5	6

$$2 \cdot 3 \cdot 6 = 36$$

De:

1	2	3
0	-1	4
0	0	5

$$1 \cdot (-1) \cdot 5 = -5$$

8. Determinante de Matriz Transposta

De:

A =

1	2	3
0	1	4
2	1	0

10. Calcule $\det(A)$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 2 \\ 2 & 1 & 0 \end{pmatrix}$$

$$1 \cdot \text{cofator}_{11} + 0 \cdot \text{cofator}_{21} + 2 \cdot \text{cofator}_{31}$$

$$\text{Memor} = 1 \cdot 0 - 2 \cdot 1$$

$$\text{Memor} = 2 \cdot 4 - 3 \cdot 1$$

$$= 0 - 2$$

$$= 8 - 3$$

$$= -2$$

$$= 5$$

$$= -2 + 10$$

$$\text{cofator} = 1 \cdot (-1)^{1+1} \cdot (-4)$$

$$\text{cofator} = 2 \cdot (-1)^{3+1} \cdot 5$$

$$= 6$$

$$= 1 \cdot (-1)^2 \cdot (-4)$$

$$= 2 \cdot (-1)^4 \cdot 5$$

$$= 1 \cdot 1 \cdot (-4)$$

$$= 2 \cdot 1 \cdot 5$$

$$= -4$$

$$= 10$$

11. Determine $\det(A^T)$

$$A^T = \begin{pmatrix} 1 & 0 & 2 \\ 2 & 1 & 1 \\ 3 & 2 & 0 \end{pmatrix}$$

$$0 \cdot \text{cofator}_{11} + 1 \cdot \text{cofator}_{21} + 4 \cdot \text{cofator}_{31}$$

$$\text{Memor} = 1 \cdot 0 - 2 \cdot 3$$

$$\text{Memor} = 1 \cdot 1 - 2 \cdot 2$$

$$= 0 - 2 \cdot 3$$

$$= 1 - 4$$

$$= -6$$

$$= -3$$

$$= -6 + 12$$

$$\text{cofator} = 1 \cdot (-1)^{2+2} \cdot (-6)$$

$$\text{cofator} = 4 \cdot (-1)^{3+2} \cdot (-3)$$

$$= 6$$

$$= 1 \cdot (-1)^4 \cdot (-6)$$

$$= 4 \cdot (-1)^5 \cdot (-3)$$

$$= 6$$

$$= 1 \cdot 1 \cdot (-6)$$

$$= 4 \cdot (-1) \cdot (-3)$$

$$= 1 \cdot (-6)$$

$$= 4 \cdot (-3)$$

$$= -6$$

$$= 12$$

12. Determinante de produto e inversa

$$\text{Dado: } \det(A) = 2$$

$$\det(B) = -3$$

$$\text{com } A, B \in \mathbb{R}^{n \times n}$$

Calcule:



$$1a. \det(A|B) = 2 \cdot (-3) = -6$$

$$1b. \det(A^2) = 2^2 = 4$$

$$1c. \det(B^{-1}) = \det(-3^{-1}) = \frac{1}{-3}$$

$$= -\frac{1}{3}$$

$$= -\frac{1}{3}$$

$$3$$

$$1d. \det(A^{-1}B) = \det(2^{-1}) = \frac{1}{2} \cdot -3$$

$$= -\frac{3}{2}$$

$$= -\frac{3}{2}$$

$$2$$

10. Propiedades

$$\det(A) = 2$$

Matrices de orden m .

determinante o valor:

$$\det(B) = -3$$

$$1a. \det(3A) = 3^m \cdot 2$$

$$1b. \det(-B) = (-1)^m \cdot (-3)$$

$$1c. \det(2AB) = 2^m \cdot 2 \cdot -3$$

$$= 2^m \cdot -6$$

$$1d. \det(A^{-1}B) = \frac{1}{2} \cdot -3 = -\frac{3}{2} = -\frac{3}{2}$$

11. Propiedades de $\det(A) = 2$

$\det(B) = -3$, donde A y B son matrices de orden m .

Calcular:

1a. $\det(AB) = 2(-3) = -6$

1b. $\det(3A) = 3^3 \cdot 2$

1c. $\det(A^T) = 2$

1d. $\det(B^{-1}) = \frac{1}{-3} = -\frac{1}{3}$

1e. $\det(A^2) = 2^2 = 4$

1f. $\det(AB^T) = 2 \cdot (-3) = -6$

m. matriks sh. singular

$B = (A)$ jeb.

$E = (B)$ jeb.

n. matriks sh. nonsingular

l. matriks a. singular

$B \cdot E = (AE)$ jeb.

$(B^{-1})^{-1} (I) = (B^{-1})$ jeb.

$B \cdot E^{-1} = (BAE)$ jeb.

$B \cdot E^{-1} =$

$B \cdot E^{-1} = E^{-1} \cdot I = (B^{-1}A)$ jeb.

$B \cdot E^{-1} =$

$B = (A)$ jeb.

m. matriks sh. singular, antr. B u A sh. sh. $E = (B)$ jeb.

